

Yutaro Kiyota

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EDUCATION

Misooouri University of Science and Technology

Rolla, MO

Bachelor of Science in Computer Science, Minor in Computer Engineering

Aug. 2023 – May 2027

PROJECTS

Real-Time Flight Avionics System | C, Zephyr RTOS, Raspberry Pi Pico, SPI, IMU

- Built a real-time flight avionics prototype in C and Zephyr RTOS on a Raspberry Pi Pico, successfully integrating sensor acquisition, processing, telemetry, and system health monitoring into a single embedded system
- Designed a 100 Hz IMU pipeline over SPI with a thread-safe ring buffer for 12-byte accelerometer and gyroscope samples, enabling continuous sensor acquisition and processing across concurrent threads
- Developed a 10 Hz UART telemetry system that converts high-frequency sensor data into timestamped measurements for a ground monitoring application while maintaining onboard 100 Hz processing
- Implemented mutex-protected shared data, watchdog-based recovery, and health monitoring to create a fault-tolerant architecture capable of continuing operation during software or sensor faults

Embedded Spatial Audio Calibration System | C, Zephyr RTOS, Python, Raspberry Pi Pico

- Built an embedded audio calibration system in C and Zephyr RTOS that successfully measures speaker timing differences and calculates corrections for synchronized multi-speaker audio playback
- Integrated an I2C OLED display, ToF distance sensor, analog potentiometer, GPIO controls, and USB CDC communication, enabling interactive configuration and calibration directly from the embedded device
- Developed a JSON-based serial protocol between firmware and a Python desktop application for speaker registration, device configuration, and calibration data logging, creating an end-to-end calibration workflow
- Implemented chirp-based signal processing with Python and NumPy to measure speaker latency and calculate per-speaker delay compensation for improved synchronization across multiple speakers

Stock Prediction Platform | Python, TypeScript, FastAPI, SQL, React, XGBoost

- Built a full-stack stock prediction platform combining historical market data with financial news sentiment to predict short-term stock price movement with 58% accuracy using time-series and sentiment features
- Designed a relational SQL database storing historical stock prices, financial news, sentiment scores, model predictions, and actual results with indexed queries for efficient data retrieval
- Developed a FastAPI backend with scheduled prediction and market-data processing and connected it to a TypeScript/React dashboard for interactive prediction and performance visualization

EXPERIENCE

Software Developer – Satellite Research Team

Aug. 2023 – Present

Missouri University of Science & Technology

Rolla, MO

- Designed and extended an Operating System Abstraction Layer over FreeRTOS to standardize task scheduling and file system interfaces across CubeSat flight software components
- Designed and debugged a multithreaded C++ Ethernet communication module supporting concurrent satellite-to-ground and satellite-to-satellite data exchange, including GDB-based debugging of flight software communication issues

Senior Resident Assistant / Resident Assistant

Aug. 2024 – May 2026

Missouri University of Science & Technology

Rolla, MO

- Supported and mentored residents while managing community concerns, enforcing housing policies, and responding to student needs in a residential community
- Coordinated programs and community activities while collaborating with housing staff to address conflicts, maintain a safe living environment, and support resident well-being

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, TypeScript, HTML/CSS **Frameworks/RTOS:** Zephyr RTOS, React, FastAPI
Embedded: SPI, I2C, UART, ADC, GPIO, USB CDC, ARM Cortex-M **Developer Tools:** Git, Docker, VS Code, Visual Studio, GDB, CMake, West **Libraries:** NumPy, pandas, Matplotlib